CAPSTONE PROJECT SUBMISSION TEMPLATE

Template 1: Ideation Phase

Project Title:	**OrderPulse: Food Order Performance Analytics**	**Template Phase:**	**Template 1: Ideation Phase**
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GitHub Repo:	https://github.com/ArnaovGarg925/OrderPulse-Food-Order-Performance-Analytics	**Demo Video:**	Google Drive (1080p Video Live)

1. Project Title & Domain

Title: OrderPulse: Food Order Performance Analytics

Domain: Cloud Data Warehousing, E-Commerce Analytics & Quick-Commerce Logistics

Target Platform: Snowflake Cloud Data Platform & Streamlit Web Applications

2. Problem Statement

Modern online food ordering platforms operate at hyper-scale, capturing millions of data points across diners, restaurant partners, multi-course menus, line-item baskets, delivery agents, and financial settlements. However, in transactional operational environments, this rich telemetry is isolated across disparate silos (e.g., MySQL ordering databases, restaurant partner portals, customer CRM records, and third-party delivery dispatchers).

Because these disparate sources are not integrated into an unified analytical model, business decision-makers face severe challenges: (1) Discrepancy between Gross Merchandise Value (GMV) and realized net revenue due to unclassified delivery cancellations; (2) Blind spots in customer retention, making it impossible to separate one-time buyers from high-value repeat loyalists; (3) Subjective partner negotiations lacking empirical evidence connecting restaurant star ratings with customer order volume; and (4) Delivery SLA compliance failures during meal peak windows that go undetected until customer churn occurs.

3. Proposed Solution & Architecture Overview

OrderPulse solves this fragmentation by implementing an enterprise-grade cloud data analytics platform powered by **Snowflake's Medallion Data Architecture** (RAW ➔ STAGING ➔ CURATED):
• **RAW Layer:** Automated ingestion of operational CSV dumps into schema-permissive landing tables via internal stages.
• **STAGING Layer:** Rigorous data cleansing, strict type-casting, string hygiene, and financial outcome classification.
• **CURATED Layer:** High-performance dimensional Star Schema (2 Fact Tables and 5 Dimension Tables) with surrogate key hashing, complemented by persona-driven analytical data marts for instant consumption.
• **Serving Layer:** An interactive 5-tab **Streamlit Web Application** providing real-time KPI scorecards, dynamic filtering, and interactive Plotly visualizations for stakeholders.

4. Core Objectives

- **Single Source of Truth:** Ingest and reconcile 12,000 orders and 32,921 line items across 7 metropolitan Indian cities.
- **Dimensional Analytics:** Design and implement surrogate-key dimensional models with RFM customer loyalty tiers and rating brackets.
- **Executive Financial Reconciliation:** Clearly differentiate between placed GMV (■96.22 Lakhs) and realized net revenue (■89.38 Lakhs).
- **Persona-Based Decision Support:** Equip Priya (Customer Engagement), Daniel (Restaurant Partnerships), and Grace (Executive Strategy) with tailored analytical marts.

- **Performance Benchmarking:** Achieve sub-second query latency and demonstrate over 6x query performance acceleration using micro-partition pruning.

5. Stakeholder Personas & Value Proposition

Persona	Role & Scope	Key Business Question Solved
Priya	Customer Engagement Manager	What is our repeat vs. single-order customer ratio? How does basket item count affect overall order value?
Daniel	Restaurant Partnerships Manager	Do higher star ratings directly drive order volume? Which cuisines and menu items generate peak platform GMV?
Grace	Business Executive	What is our monthly revenue trajectory? What percentage of revenue is lost to cancellations and where are delivery SLAs failing?